

NEXUS GUILDS

AI Foundational

From Professional to AI Engineer

Week 4: The Interface — APIs, Automation & Integration

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Pre-Reading: ~1.5 Hours | In-Class Session: 2 Hours | 75% Practical

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Welcome to Module 2: The Workshop

You are now an AI user. Module 2 takes you from user to AI integrator — connecting AI to real workflows through APIs, automation, and agents.



The 15 Automations Challenge

Andrés's team has 15 manual processes to automate: classifying emails, summarising meetings, screening applications, drafting posts, routing tickets. The problem? Wildly different skill levels — 3 code Python fluently, 5 have never written a line.

His boss says: "Find a strategy that works for everyone." Tonight, Andrés discovers automation is a spectrum, not a single skill.



ACTION REQUIRED: Create free Zapier account (zapier.com) + free Make account (make.com) + have Gmail/Google account ready.

Week 4 Overview



Part 1: Pre-Reading (~1.5h)

- What Is an API?
- Five Integration Patterns
- The Tool Spectrum (4 tiers)
- Anatomy of an AI Automation



Part 2: In-Class Session (2h)

- 4.A: Three Ways to Automate (demo)
- 4.B: Build Your First Automation
- 4.C: Process Mapping Workshop
- Preview: Week 5 agents & orchestration

75% Practical

PART 1

APIs and the Automation Spectrum

Complete before your in-class session | Estimated time: ~1.5 hours

What Is an API?

Reading 1 | ~25 minutes



The Restaurant Analogy

You (client) give your order to the waiter (API). The waiter takes it to the kitchen (server/model). The kitchen prepares the food and the waiter brings it back (response). You never go into the kitchen yourself.

When you call an AI model's API: your prompt is the order, the AI model is the kitchen, and the API is the waiter.

The Four API Concepts



Request / Response

Send prompt, get output back



Authentication

API key proves who you are



Rate Limits

Max requests per minute/hour



Endpoints

Different URLs for different capabilities

Five Integration Patterns

Reading 2 | ~20 minutes

Synchronous

Send request, wait, proceed

Form submit → API → result on page

Event-Driven

Trigger fires → background processing

New email → classify → route

Batch

Collect items, process together on schedule

Nightly: process all day's tickets

Streaming

Receive partial results as generated

ChatGPT tokens appearing word by word

Tool Use

AI decides which tools to invoke mid-response

Claude calls weather API + calculator

Start with Event-Driven: trigger fires → AI processes → result delivered. This is what Zapier and Make are built for.

The Tool Spectrum

Reading 3 | ~25 minutes

1



No-Code

Zapier, Make

Simple triggers, linear flows

Maint: Low

2



Low-Code

Retool, n8n

Custom logic, branching, DB connections

Maint: Medium

3



AI-Assisted Code

Cursor + API, Replit + API

Full flexibility, complex business logic

Maint: Med-High

4



Full Code

Python SDK, Node.js SDK

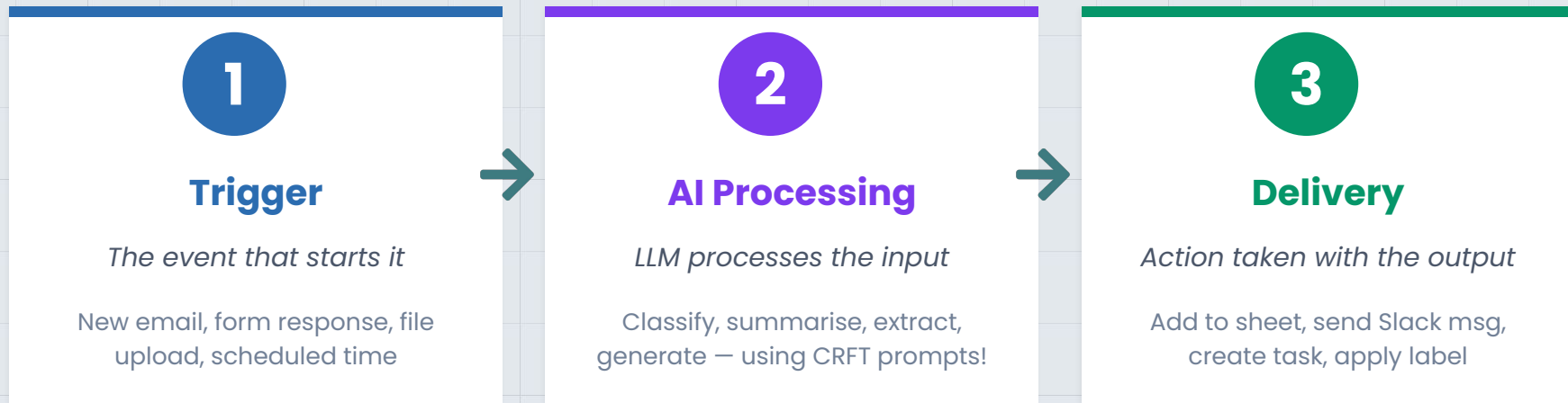
Unlimited — production systems, scale

Maint: High

80% of automation needs can be solved at No-Code. Reserve code for the 20% that genuinely needs it.

Anatomy of an AI Automation

Reading 4 | ~20 minutes



Email Classification Example

Trigger: new email in Gmail → AI: "Classify as Urgent, Question, FYI, or Spam. Respond with only the category name." → Delivery: apply Gmail label + optional forward.

PART 2

In-Class: The Interface

APIs, Automation & Integration | 2 Hours | 75% Practical

Session Plan

0:00–0:10

Opening: "The 15 Automations" story + Self-Study 2 debrief

Conceptual

0:10–0:25

Core Teaching: API fundamentals, Tool Spectrum, integration patterns

Conceptual

0:25–0:50

Exercise 4.A: Three Ways to Automate (Zapier / Make / cURL)

Practical

0:50–1:30

Exercise 4.B: Build Your First Automation (trigger → AI → delivery)

Practical

1:30–1:50

Exercise 4.C: Process Mapping Workshop (5–10 processes)

Practical

1:50–2:00

Preview: Week 5 agents and multi-step orchestration

Conceptual

Exercise 4.A: Three Ways to Automate

25 minutes | Same automation, 3 implementations | Follow along

New Gmail email → LLM classifies as Urgent / Question / FYI / Spam → Apply Gmail label

Zapier

Visual builder: drag-and-drop trigger, AI action, label action.

Fastest to build. No code. Limited flexibility.

Make

Visual flow with HTTP module calling the API directly. You see the raw request.

More control. You see the API call underneath.

cURL / Python

Terminal command and script calling the same API with the same prompt.

Maximum control. You handle everything yourself.

Key question: Which version would you maintain? Which could your non-technical colleague build?

Exercise 4.B: Build Your Automation

40 minutes | Zapier or Make | Trigger → AI → Delivery

1

Email Analyser

Trigger:

Email

AI Step:

Resume

Delivery:

Slack Message

2

Meeting Summariser

Trigger:

New file in Drive

AI Step:

Summarise notes

Delivery:

Send to Slack

3

Application Screener

Trigger:

Email with "Application"

AI Step:

Extract + rate fit

Delivery:

Add to Sheet

4

Planner

Trigger:

New PRD

AI Step:

Generate atomic user stories

Delivery:

An Epic in Jira

5

Ticket Router

Trigger:

Form / email

AI Step:

Categorise + urgency

Delivery:

Create task + priority



Start simple. One trigger, one AI step, one delivery. A working simple automation beats an incomplete complex one.

Exercise 4.C: Process Mapping

20 minutes | In pairs | Map 5-10 processes to Tool Spectrum tiers

Process	Volume	Tier	Justification	Effort
Classify customer emails	200/day	No-Code	Simple classification, Zapier handles it	2 hours
Generate sales proposals from CRM	20/week	Low-Code	Needs CRM integration + branching logic	1 day
Analyse contracts vs compliance rules	5/week	AI-Assisted Code	Complex business logic + long documents	3 days

Document: process name, volume/frequency, recommended tier, justification, and estimated effort. This feeds directly into Week 5's agent design exercises.

How to Choose Your Tier

1. Who will build it?
2. Who will maintain it?
3. How complex is the logic?
4. What scale do you need?

Week 4 Deliverables



Working Automation

A functional Zapier or Make automation with trigger → AI step → delivery. Screenshot or share link + note on which tier and why.



Process Map

5-10 processes mapped to Tool Spectrum tiers with justification. Feeds into Week 5 agent design.

Your Portfolio: 6 Artefacts

W1: Compass

W1+: Playbook

W2: Selection Matrix

W3: Prototype

SS2: Knowledge Guide

W4: Automation + Map

Looking Ahead: Week 5



Andrés's automations are running — emails classified, meetings summarised. But they're isolated. Marketing's content pipeline doesn't talk to Sales' lead qualification. Individual automations are like individual musicians. Andrés needs an orchestra.

Next week: chaining AI steps into multi-step workflows, designing agents that make decisions autonomously, and mapping cross-departmental AI workflows. Week 5 is where AI integration becomes AI orchestration.

Week 5: From Automation to Orchestration

- Multi-step workflows chaining AI processes together
- Autonomous agents that make decisions and take actions
- Cross-departmental AI workflow mapping

Bring your Process Map from Exercise 4.C — it's the raw material for Week 5.

***"The expert in anything
was once a beginner."***

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Source Meridian